ITEC 235-940 System Administration

Domain Join Lab

Introduction

As an SIU student, you have access to non-trial versions of most every Microsoft offering. These licenses are intended to be used as a learning resource and are not for commercial use. For this lab, you will download and install Windows 10 Education Edition or Windows 11 Education Edition, and then join it to the domain you created in Unit 4. The instructions are written specifically for Windows 10 Education Edition, but they will work for Windows 11 Education Edition as well.

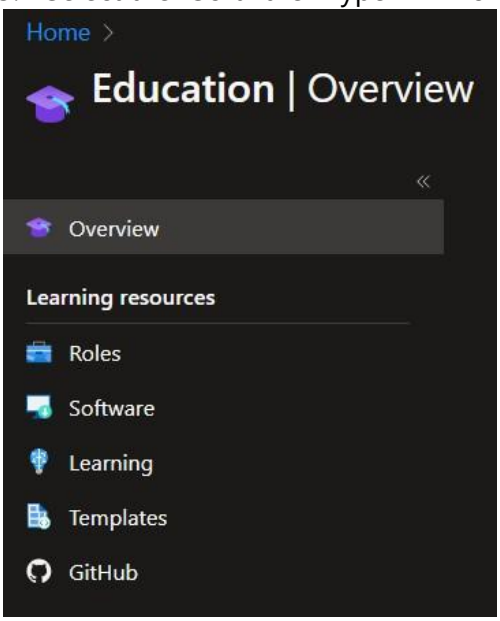
WARNING: Follow the steps sequentially. The entire installation process should take place within a single window (the virtualized computer). Your physical computer should never reboot. If you do not follow these steps, you could install Windows 10 Education on your home computer and lose all of your data. I recommend completing a backup before proceeding. Please do not hesitate to reach out to me if you are unsure of something.

Download and Install Windows 10 Education Edition

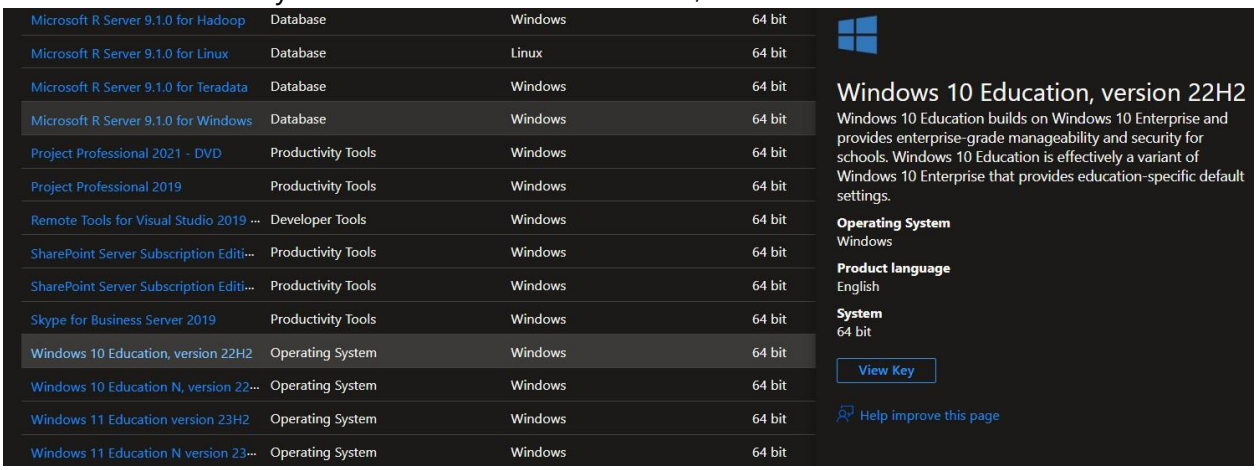
1. Open a web browser and access the following URL: https://azureforeducation.microsoft.com/devtools

2. Select the "Sign In" hyperlink and use your SIU credentials to login.

3. Select the "Software" hyperlink from the navigation column on the left.



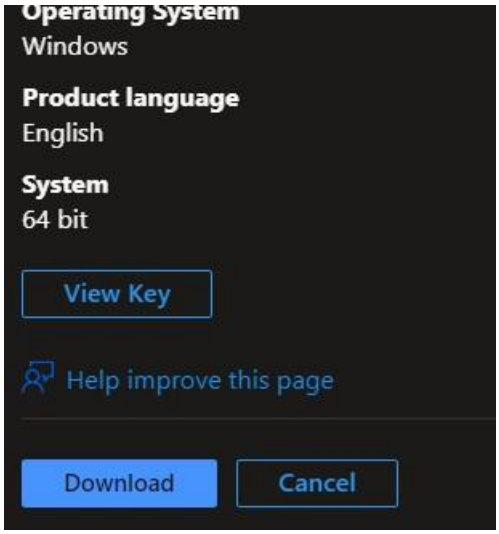
4. Scroll down until you find “Windows 10 Education, version 22H2”.



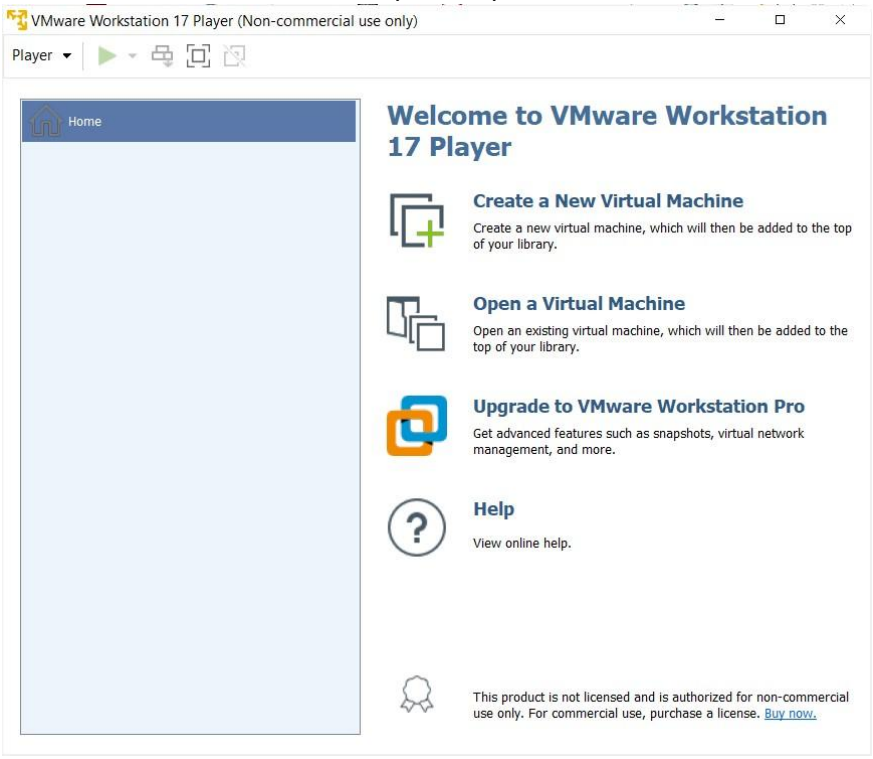
5. Click on the “Windows 10 Education, version 22H2” hyperlink and select the blue Download button that appears in the left column. Note: Download is not the same as install. For this step, you are simply going to save the file to your hard drive. I suggest the Downloads folder. **Do not double click the downloaded iso file.**



6. Select the “View Key” button and make note of the product key. You will need this during the installation process.

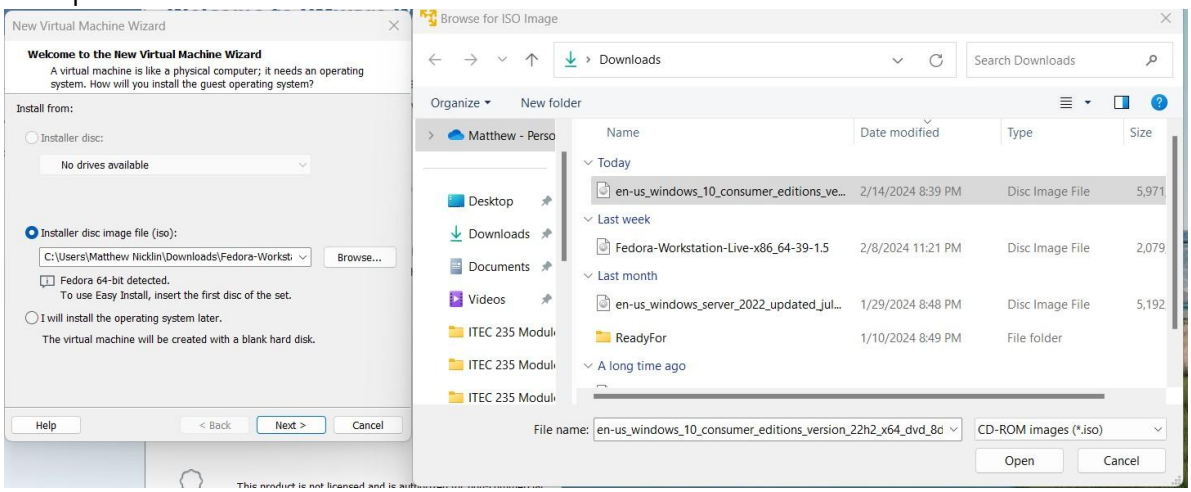


7. Once the download is complete, open VMWare Workstation.

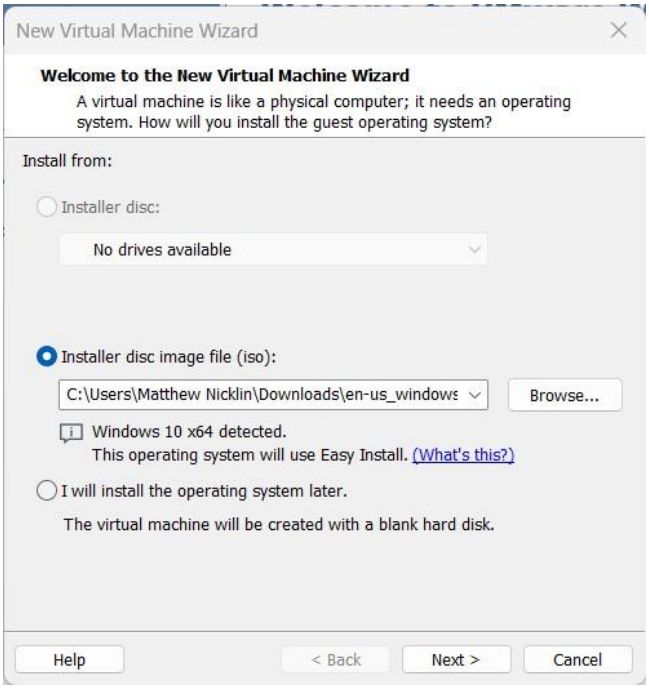


8. Select “Create a New Virtual Machine”.

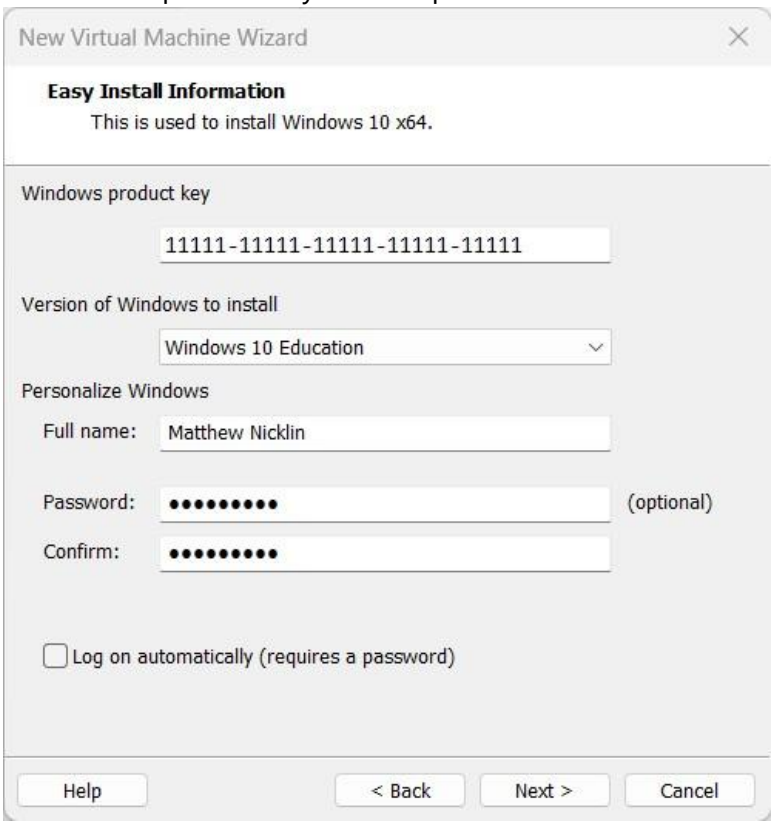
9. Select the Browse button and select the iso file downloaded in step 5 and select the Open button.



10. Select the Next button.



11. Enter the product key from step 6.



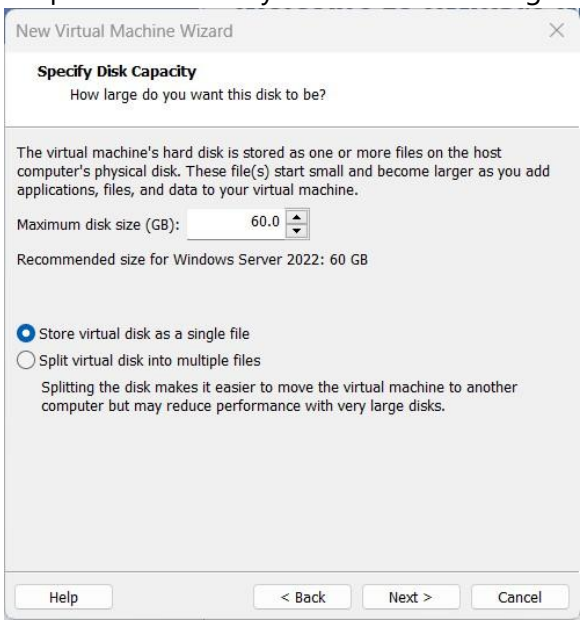
12. Select “Windows 10 Education” from the Version of Windows to install drop down.

13. Enter a password that is at least eight characters in length, includes one uppercase letter, one lowercase letter, one number, and one symbol. I like to use the password ChangeMe1!

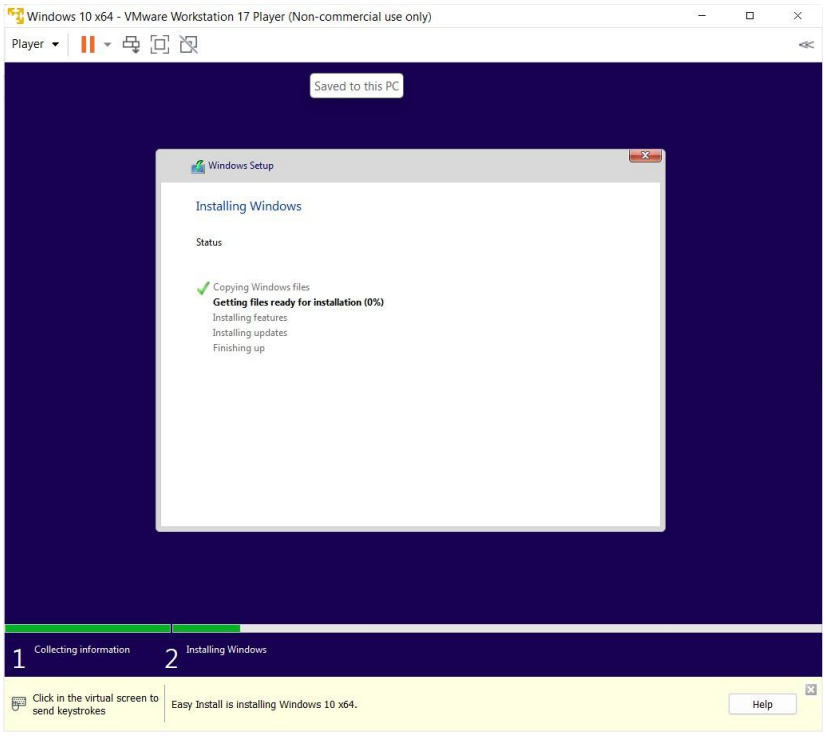
14. Select the Next button.

15. Change the name of the virtual machine and/or location if desired and select the Next button.

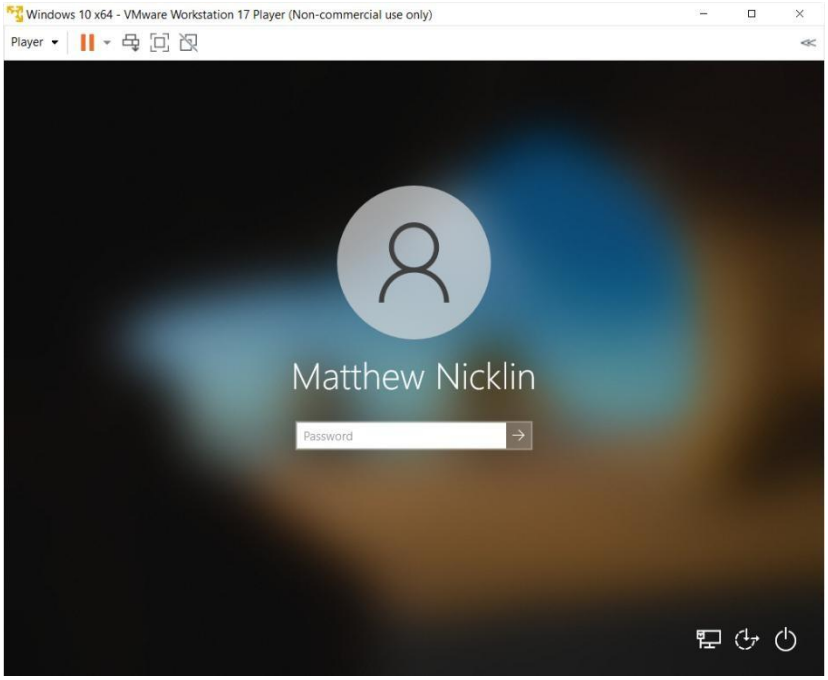
16. I prefer to save my virtual disk as a single file. You can make this choice if you prefer.



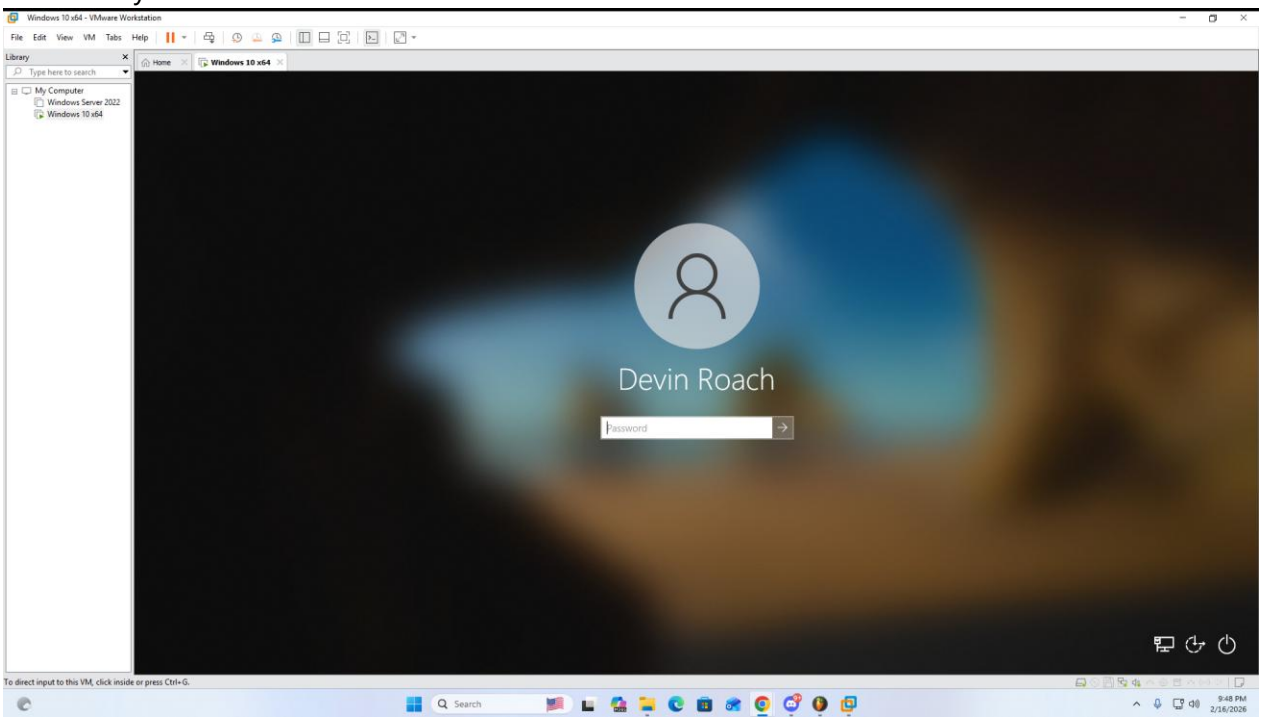
17. Select the Next button.
18. Select Finish to begin the installation process.



19. Upload a screen capture of your Windows Server VM to the Domain Join Lab submission box.



My Answer:

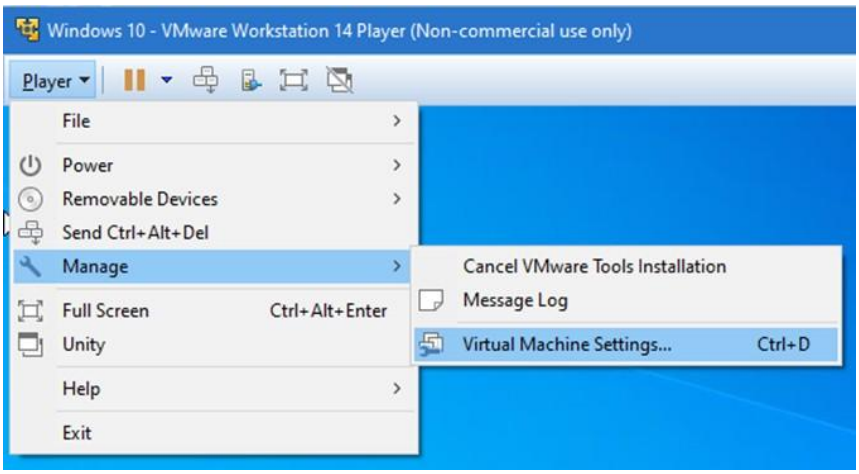


Join Windows 10 Education to Server 2022 Domain

Now that Windows 10 Education Edition has been installed, it can be joined to the domain. To begin this process, it is necessary to start the Windows Server 2022 VM.

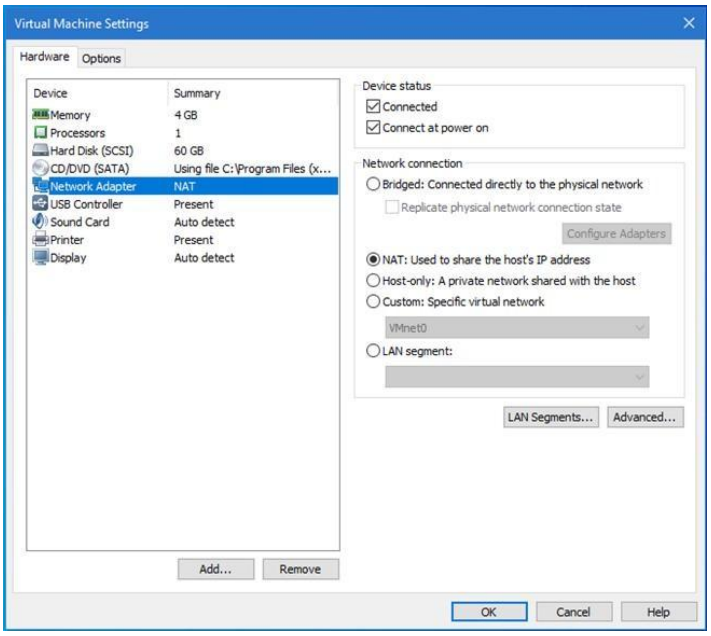
1. You must first assign a static IP address to the Windows Server 2022 VM. Before assigning a static IP address, it is recommended that you verify that your VM is using NAT. To do this, select Player > Manage > Virtual Machine Settings from the upper left corner of the Windows Server 2022 VM window.

Note: You can use different types of connections, but to keep things simple, it is recommended that you follow the provided instructions. If you are a VMware pro and want to assign a static IP a different way, please feel free to do so.



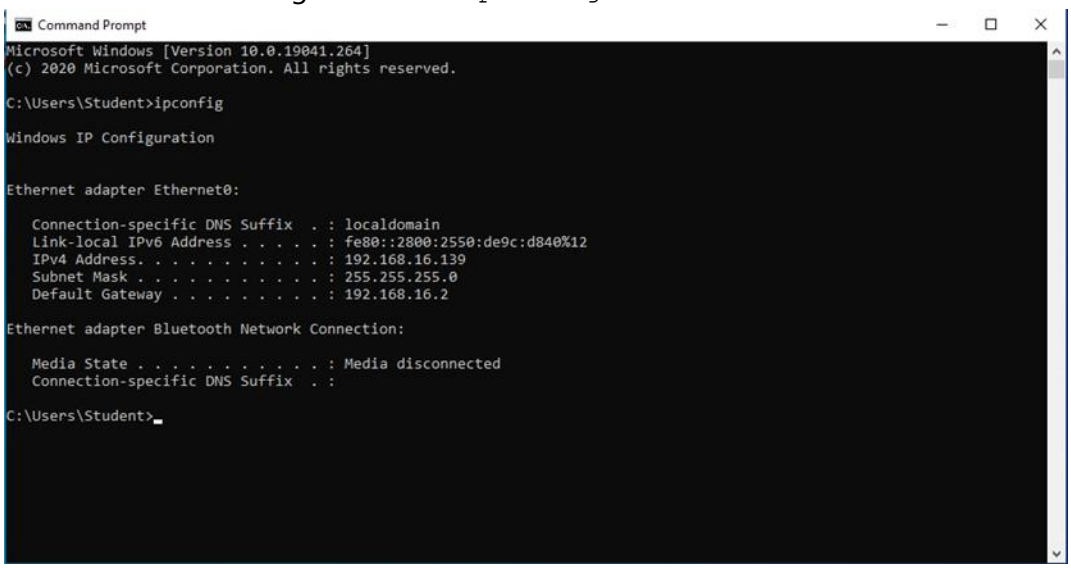
2. Click Network Adapter in the left column and verify that NAT is chosen under Network connection. If it is not, please make the change and click the OK button. Reboot the server if a change is made.

By default, VMware assigns an IP address from the 192.168.x.x range. The first address, 192.168.x.1 is used to identify the virtual host adapter. The second address, 192.168.x.2 is used to identify the NAT gateway address.

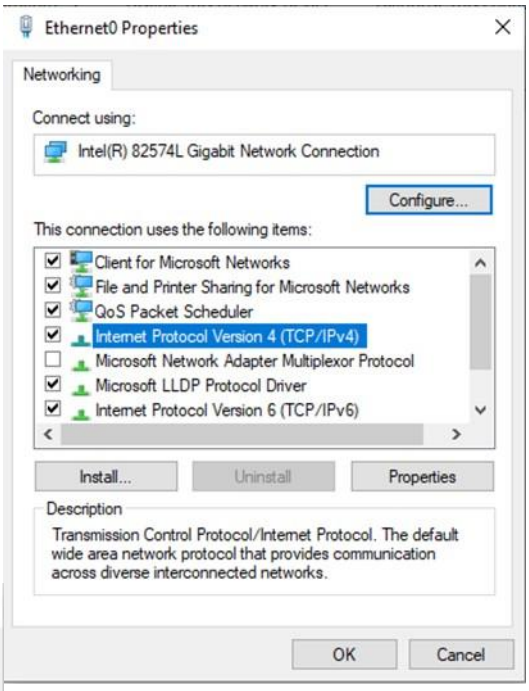


3. After verifying the IP configuration for the server, verify the configuration of the Windows 10 workstation.

4. Once the workstation and server have been configured with the same network adapter (NAT), open a command prompt on the Microsoft Server 2022 VM and enter the following command: `ipconfig /all`



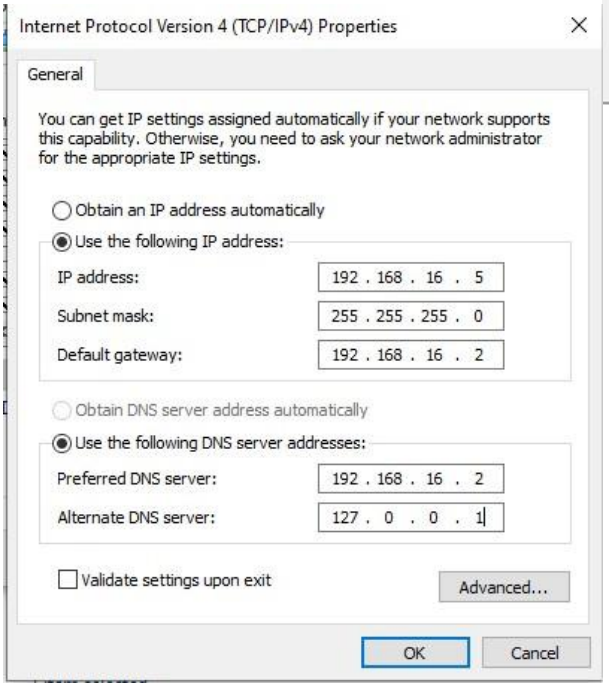
Note (as in write down) the IP address and default gateway address. In this example, the IP address is 192.168.16.139 and the default gateway is 192.168.16.2. Remember that VMware uses a DHCP range of 128 – 254. Therefore, we simply need to assign the server an IP address from outside of that range.



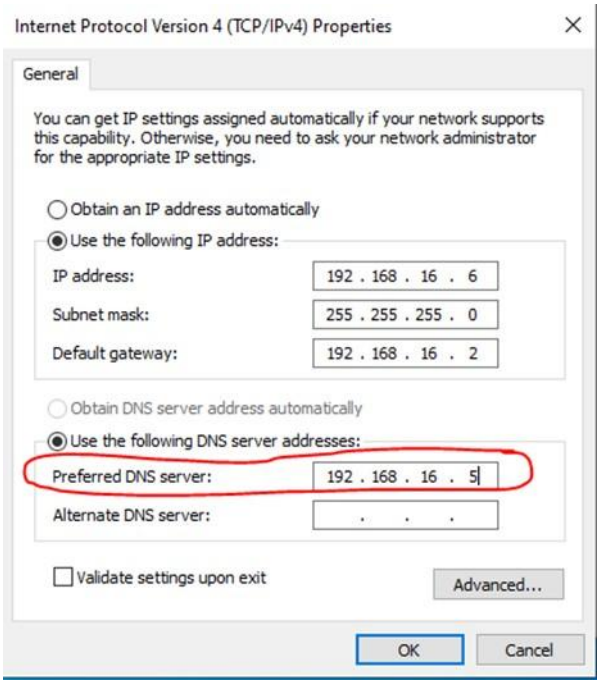
5. On the Server 2022 VM (*click the desktop one time to make sure you have selected it*), hold down the Windows key and tap the R key. This will open the Run command window. Enter ncpa.cpl in the Run text box and click the OK button.

6. Right click on Ethernet0 connection (assuming the default) and select Properties.

7. Click "Internet Protocol Version 4 (TCP/IPv4)" to highlight it and click the Properties button.

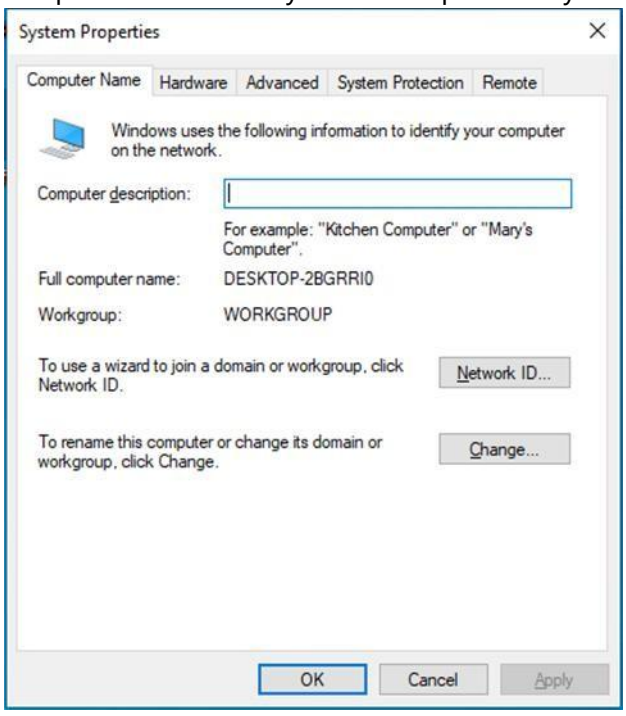


8. Enter the IP address of your choosing (I used 192.168.16.5). The first three octets of the IP address you select will need to match the first three octets of the IP address noted in Step 4. My first three octets are 192.168.16. The final octet can be a number of your choosing 3 – 127. See narrative from Step 2 for additional information. The Subnet Mask is 255.255.255.0. The first three octets of the Default Gateway and Preferred DNS server will match the first three octets of the IP address noted in Step 4. For good measure enter 127.0.0.1 in the Alternate DNS server field.



9. Select the Windows 10 Education VM and repeat steps 5 – 8 to configure a static IP address on your workstation. To keep it simple, you can increment the IP address you used for your server by 1. For example, since I used 192.168.16.5 for my server, I assigned 192.168.16.6 to my workstation. In the Preferred DNS server field, enter the IP address of your Server 2022 VM. For example, since I used 192.168.16.5 for my server, I entered 192.168.16.5 in the Preferred DNS server field for my workstation.

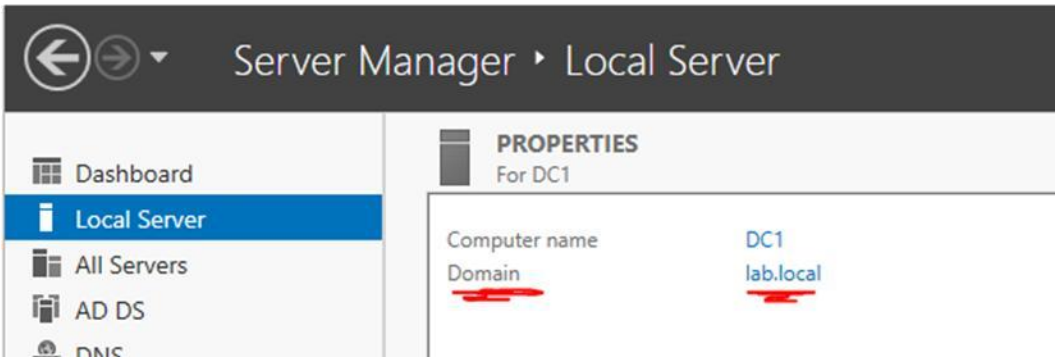
10. From the Windows 10 Education VM, type sysdm.cpl in the *Type here to search* field and press the Enter key. This will open the System Properties window.



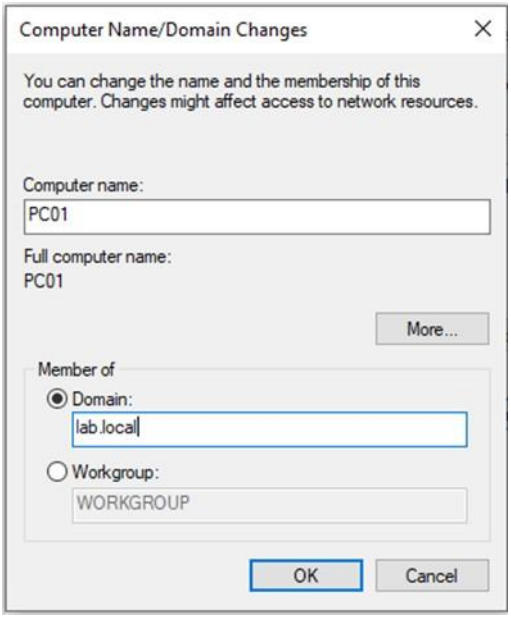
11. Click the Change button found in the lower right side of the System Properties window.

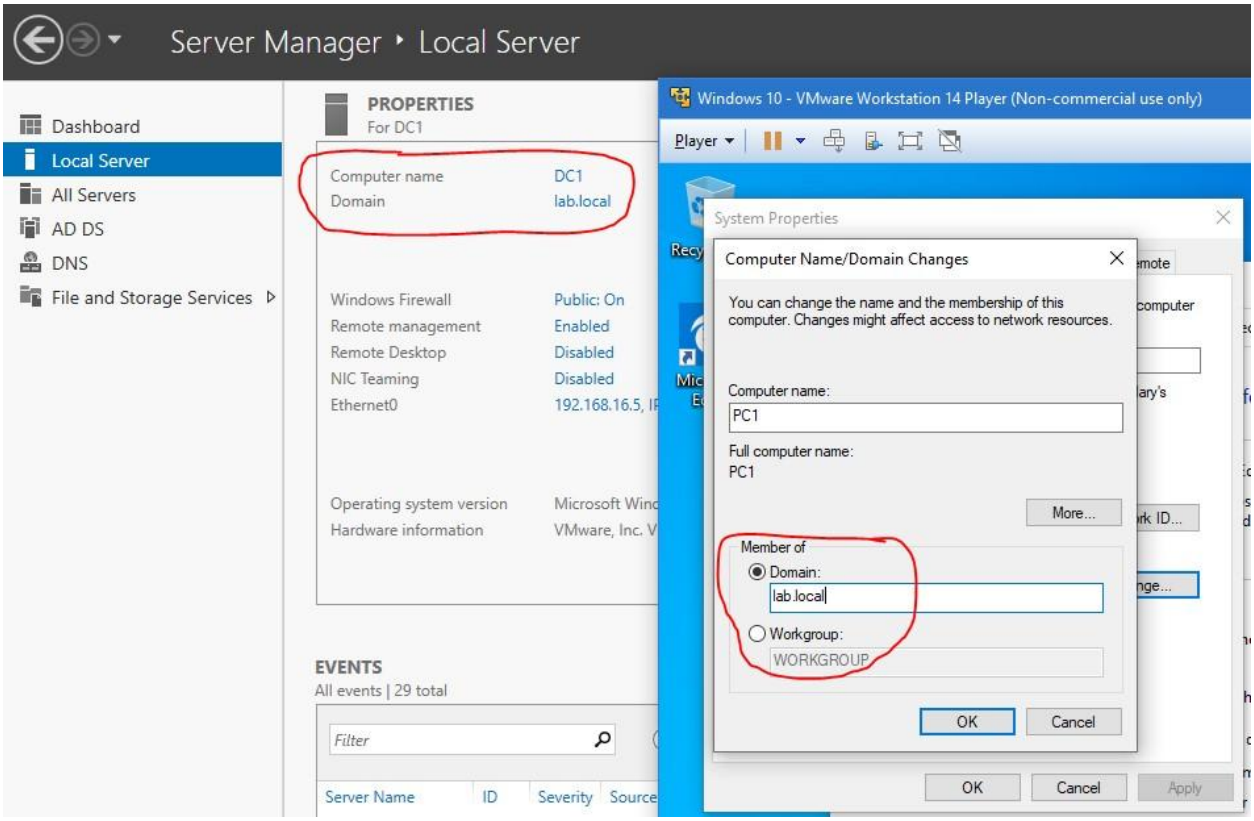
12. Rename the computer to something explanatory. For example, I used PC01.

13. Return to the Windows Server 2022 VM. Select Local Server in Server Manager and note the name of the domain.



14. Return to the Windows 10 Education VM. Click the radio button to the left of Domain. Enter the name of the domain in the Domain: field and click on the OK button.



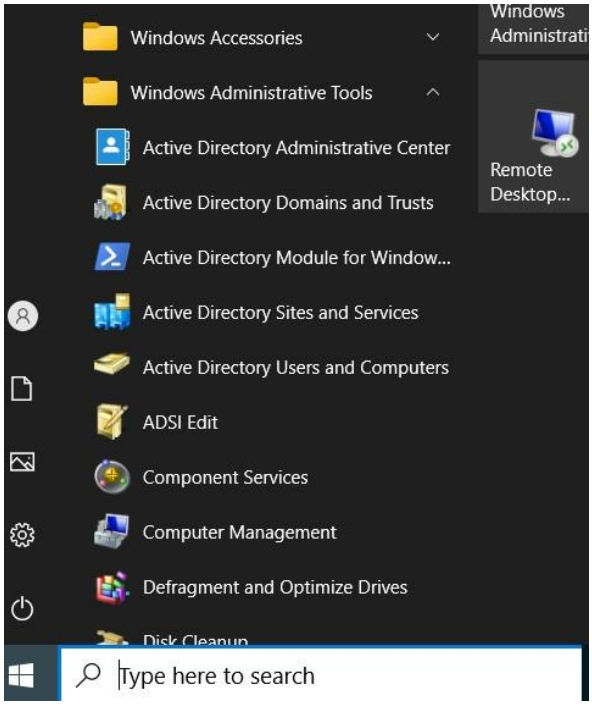


15. Enter the Username and Password you use to login to your Server 2022 VM (the domain administrator account) and click the OK button.

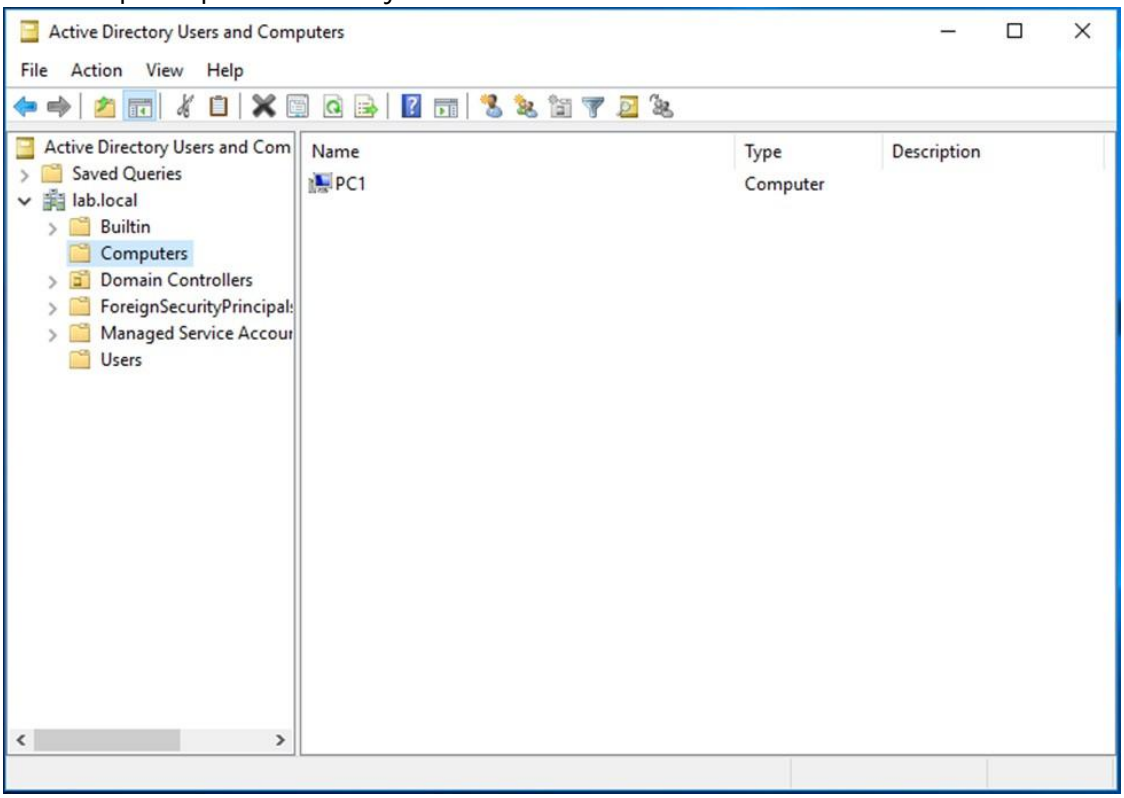
16. Hopefully, you receive a Welcome message. Click the OK button and reboot the Windows 10 Education VM.

17. You can now login to the Windows 10 Education VM using a domain user account (the username and password you use to login to the Server).

18. From the Server 2022 VM, click on the Start menu. Select Windows Administrative Tools and then Active Directory Users and Computers.



19. Select the Computers OU. Submit a screen capture of the Computers OU showing your desktop computer as an object to the Domain Join Lab submission link.



My Answer:

